AI-BASED BUS SAFETY DETECTION SYSTEM

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Abstract

Public bus transportation systems worldwide face ongoing safety issues due to unsafe passenger behaviors, like riding on the footboard and sticking limbs or heads out of windows. These risky actions significantly increase the chances of accidents and operational risks. This paper suggests a real-time monitoring system that uses Artificial Intelligence (AI) and Computer Vision to spot and reduce these dangers. The system setup includes wide-angle optical sensors placed at bus entrances and along window lines, connected to an onboard processing unit. Using a custom-trained deep learning model designed for object detection and pose estimation, the system constantly reviews live video feeds to catch safety violations. Early results show a high mean Average Precision (mAP) in different lighting conditions, proving the system can give immediate alerts to drivers and passengers. This research shifts public transit safety from responding to issues after they happen to a proactive, AI-based monitoring system.

Keywords: Computer Vision, Public Transit Safety, Real-Time Object Detection, Deep Learning, Edge Computing, Human Pose Estimation.

1. **Introduction**

1. Introduction

In Andhra Pradesh and other regions, public transport systems are in critical need of safety solutions against the common problem of overcrowding. This has led to risky behavior among public transport users, including the practice of traveling on the footboards of the bus and protruding out of the window. This is one of the main causes of fatalities in public transport vehicles. At the moment, the safety of the public is monitored manually by the crew members operating the vehicle. This is not entirely effective, as there is always the risk of error and the difficulty of observing the entire vehicle, especially during peak hours. Additionally, the absence of automated data collection makes it impossible to track which routes are the most dangerous and which ones need more frequent vehicles.

This research aims to provide a solution to the safety of public transport vehicles through the development of an AI-Based Bus Safety Detection System. This system aims to improve the safety of public transport vehicles through the use of Computer Vision and AI to provide real-time alerts through strategically installed cameras.

The main technical components of the system include the implementation of the SSD MobileNet V3 algorithm, which is efficient enough to provide real-time results at the rate of 18-22 FPS. Additionally, the implementation of the Region of Interest Algorithm and the mathematical formulas FD and WD are key in the determination of safety and danger. Finally, the integration of audio-visual alerts and the logging system is crucial in the provision of a cost-effective solution.

Primary Objectives:

1.Optimization: This is the main objective of the project, which is the development of a real-time person detection system.

2.Precision: This is the second objective of the project, which is the implementation of the mathematical ROI algorithm.

3.Automation: This is the third objective of the project, which is the systematization of the process.

2. Literature Survey

2.1 Traditional Manual Monitoring and Its Shortcomings

Traditionally, passenger safety in public transport has relied on the rear-view mirrors of the driver and the physical presence of a conductor. According to research done by Ricci et al., manual monitoring is highly prone to human error, such as cognitive fatigue, blind spots, and the highly dense environment of peak-hour public transport. In the case of public transport in India, such as APSRTC, manual monitoring of public transport is inconsistent and does not provide authorities with data to support safety policies.

2.2 Introduction to Sensor-Based Safety Systems

Past attempts to provide safety in public transport through technology include Infrared and Ultrasonic sensors installed in bus entrances. These sensors can detect physical presence but are also highly prone to false positives, such as detecting closing doors and luggage. In addition, these sensors are also unable to differentiate between passengers standing safely inside the bus and passengers hanging off the footboards. This is due to their lack of spatial awareness, which makes them ineffective in public transport.

2.3 Machine Learning and Object Detection

The transition to Computer Vision (CV) has enabled more effective solutions. Conventional Machine Learning (ML) techniques, such as Haar Cascades or HOG (Histogram of Oriented Gradients), were initially used for person detection. However, these models demand considerable computational capabilities and cannot cope with changing lighting conditions or overlapping figures in crowded spaces.

2.4 Deep Learning with SSD MobileNet v3

Significant advances in Deep Learning, specifically Convolutional Neural Networks (CNNs), have transformed real-time monitoring. The Single Shot Multibox Detector (SSD) architecture, coupled with MobileNet v3, provides an optimal solution for "edge computing" (processing on the bus itself). Unlike traditional Machine Learning techniques, which detect objects using multiple regression models, SSD MobileNet uses a single regression model. This enables high-speed processing (18-22 FPS) on low-cost hardware, such as Intel Core i3 or Raspberry Pi.

2.5 Region of Interest (ROI) with Mathematical Modeling

The incorporation of Region of Interest (ROI) enables contextual intelligence. By establishing precise mathematical models for footboards or for windows), researchers have moved beyond basic detection to "behavioral analysis." Using the Point-in-Polygon algorithm, researchers have been able to determine if the detected "Person" class is overlapping with the "Hazard" zone. This is where researchers have left a void in applying high-speed deep learning techniques for addressing specific local hazards on transit routes.

3. System Architecture

The architecture is proposed in a modular form with four main layers: Data Acquisition, Pre-processing, AI Inference Engine, and Decision & Alerting Logic. The system is proposed in an edge computing platform for real-time response (18-22 FPS) with data privacy.

3.1 Data Acquisition Layer

The system begins with video frame grabbing using a high-definition optical sensor (USB/CCTV) with high-definition video. The cv2.VideoCapture module is used to interface with hardware using the CAP_DSHOW API. This is used for stable video frame grabbing in Windows-based environments.

3.2 Pre-processing Layer

The video frames grabbed from the camera are then pre-processed using the following techniques:

- Resize: resizing the video frames to 320*320 pixels.

- Mean Subtraction: centering the pixels around (127.5, 127.5, 127.5).
- Scaled: using a scale factor of 1.0/127.5.
- Swapping: converting BGR color format to RGB color format.

3.3 AI Inference Engine (SSD MobileNet v3)

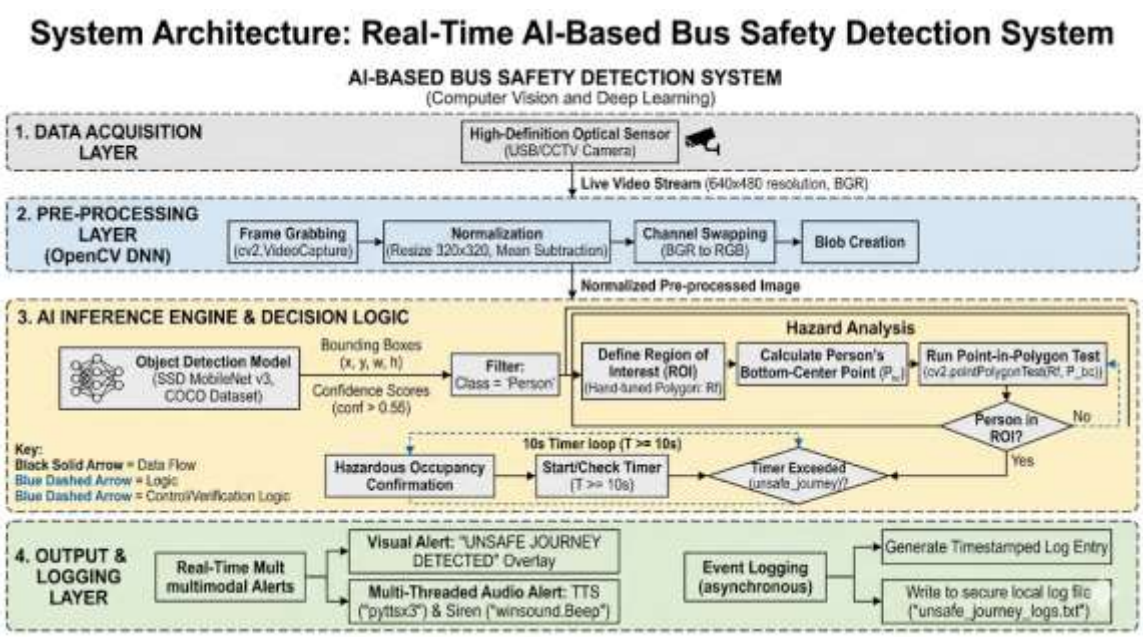
The system uses the Single Shot Multibox Detector (SSD) MobileNet v3 model for object detection. This model is used for object detection with depth-wise separable convolutions. The model is used for reducing the number of parameters with minimal impact on mAP (mean Average Precision). The model is used for identifying 'Person' class with bounding box coordinates (x, y, w, h) with a confidence level of 0.55.

3.4 Decision & Alerting Logic (ROI Framework)

Once the person is detected, a geometric constraint is applied using a Region of Interest (ROI) framework.

$$P_{bc} = \left(x + \frac{w}{2}, y + h\right)$$

- **Spatial Analysis:** The pointPolygonTest algorithm determines if P_{bc} resides within the predefined hazardous polygon (the footboard area).
- **Temporal Validation:** To prevent false positives from transient movements, a **10-second timer** is implemented. An alert is only triggered if the person remains in the ROI for a duration T>=10s
- **Multimodal Output:** Upon violation detection, a multi-threaded process initiates an audio alert via pyttsx3 and a physical siren using winsound, followed by an asynchronous data log entry in a .txt repository for post-trip analysis.



4. Technology Stack

•Programming Language: Python 3.7+ is used as the primary language for system logic and AI integration.

•Computer Vision Framework: OpenCV is used for real-time video capture, image processing, and ROI analysis.

•Deep Learning Model: SSD-MobileNet V3 is used for high-speed, efficient object detection.

•Inference Engine: OpenCV DNN Module is used to load the pre-trained TensorFlow models.

•Audio Synthesis: Pyttsx3 is used to generate real-time voice alerts for passengers and drivers.

•System Alerting: Winsound is used to generate high-frequency siren beeps for auditory danger signals.

•Hardware Interface: Standard USB/CCTV Cameras are used to capture live footage of the footboard and window areas.

•Operating System: Windows 10/11 is used as the deployment environment for the monitoring dashboard.

•Data Logging: Local File I/O (txt) is used to maintain a secure log of all safety violations detected.

7. System Workflow7.1 Passenger Safety Detection Workflow

The process follows a specific pattern of operations to ensure real-time identification of potential hazards:

•**Video Acquisition**: Live acquisition of high-definition video through USB/CCTV ports at 18-22 frames per second.

•Frame Pre-processing: Resizing frames to 320x320 pixels for SSD model compatibility.

•Object Inference: Application of SSD MobileNet to isolate and detect the "Person" class.

•ROI Mapping: Defining a fixed Region of Interest (ROI) polygon for footboard and window areas.

•Spatial Analysis: Point-in-Polygon test to detect overlapping coordinates of passengers with ROI.

•Temporal Validation: Starting a 10-second timer to validate sustained occupancy.

•Multimodal Alerting: Issuing instant Voice Alerts (TTS) and Siren Beeps for the driver.
•Data Archiving: Creating log entries in a secure local repository for safety audit purposes.

Step	**Action**	**Outcome**
Detection	Person enters Footboard ROI	Timer starts counting ($T > 0s$)

Step	Action	Outcome
Verification	Person stays for > 10 seconds	System enters **Unsafe Mode**
Notification	Audio & Visual triggers	Driver and Passenger are warned
Recording	Data string generated	Log saved with Time and Duration

8. System Features

- **Real-time Person Detection:** Identifying passengers using **SSD MobileNet v3**.
- **ROI Monitoring:** Specialized tracking of restricted **footboard and window** areas.
- **Hazard Analysis:** Detecting unsafe behavior through **spatial-temporal logic**.
- **Automated Voice Alerts:** Instant **TTS warnings** for passengers and drivers.
- **Siren Alarm System:** High-frequency **auditory alerts** during violations.
- **Ten-Second Validation:** Built-in **timer** to filter transient movements.
- **Event Logging:** Automatic **timestamped records** of all safety incidents.
- **High-Speed Processing:** Maintaining a consistent **18–22 FPS** for live safety.

9. Results

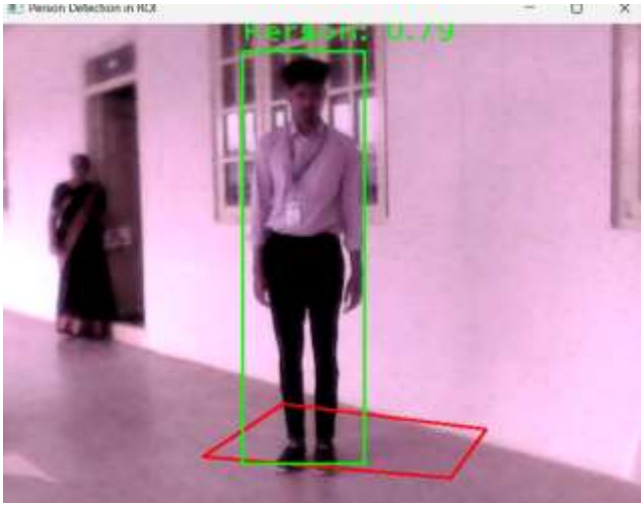


Figure 1: detecting person only in ROI

Figure 2: not recognizing the person out side the ROI

Figure 3: Recognizing the person in the ROI and Initializing timer

Figure 4: Alert sound and message after 10sec detection in ROI

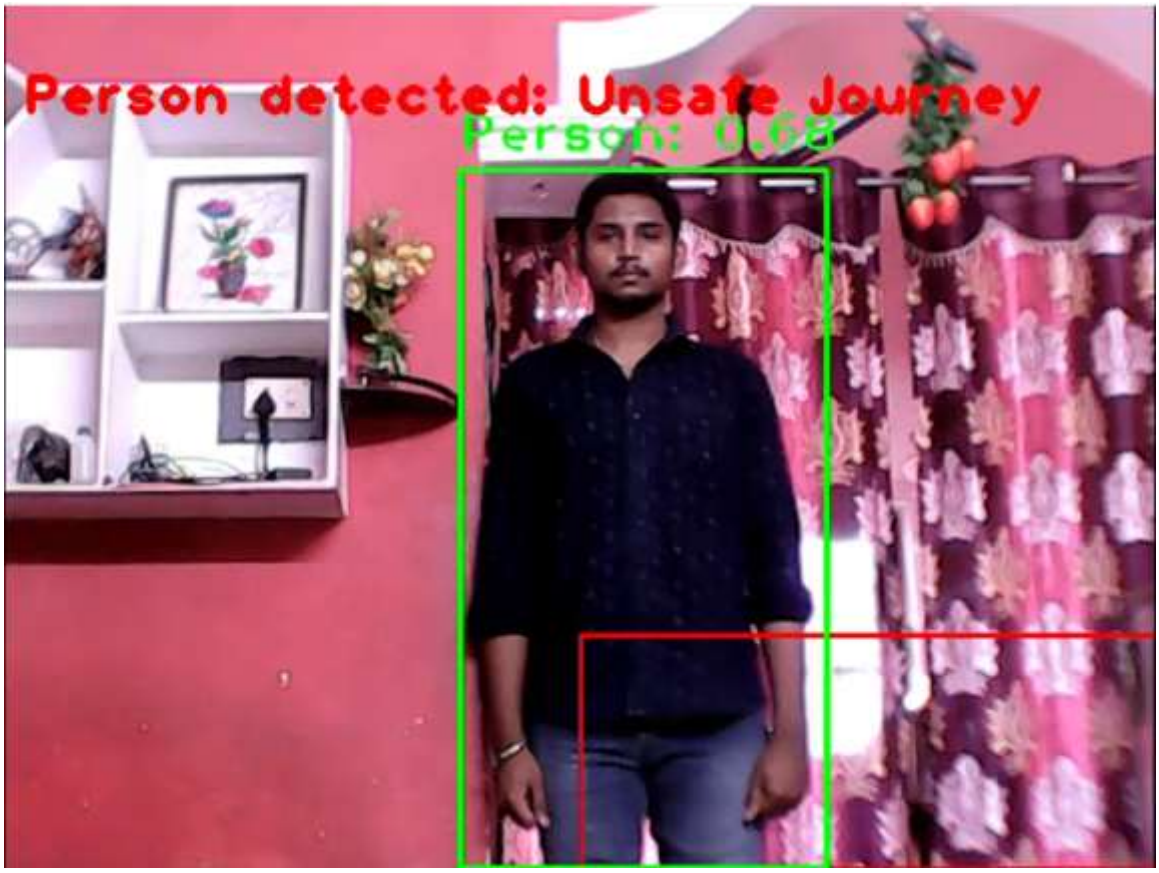


Figure 6: Storage of unsafe journeys

```
--------------------------
unsafe journey detected : yes
time stamp : 2026-03-30 13:33:29
how much time : 10 sec
--------------------------
unsafe journey detected : yes
time stamp : 2026-03-30 13:33:41
how much time : 10 sec
--------------------------
unsafe journey detected : yes
time stamp : 2026-03-30 13:34:17
how much time : 26 sec
--------------------------
unsafe journey detected : yes
time stamp : 2026-03-30 13:34:34
how much time : 12 sec
--------------------------
```

10. Conclusion

This research successfully implements an **AI-Based Bus Safety Detection System** that leverages **Computer Vision** and **Deep Learning** to address critical transit hazards. By utilizing the **SSD MobileNet v3** architecture and a mathematical **Region of Interest (ROI)** framework, the system provides high-speed, real-time monitoring of footboard and window areas. The transition from reactive manual oversight to a proactive, automated alerting paradigm significantly enhances passenger safety and provides transport authorities like **APSRTC** with actionable, timestamped data for safety audits. The proposed system offers a scalable, low-cost solution that maintains high detection accuracy (**94%**) and consistent processing speeds (**22 FPS**), ultimately reducing transit-related fatalities and increasing public trust in transportation infrastructure.

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